Supplementary

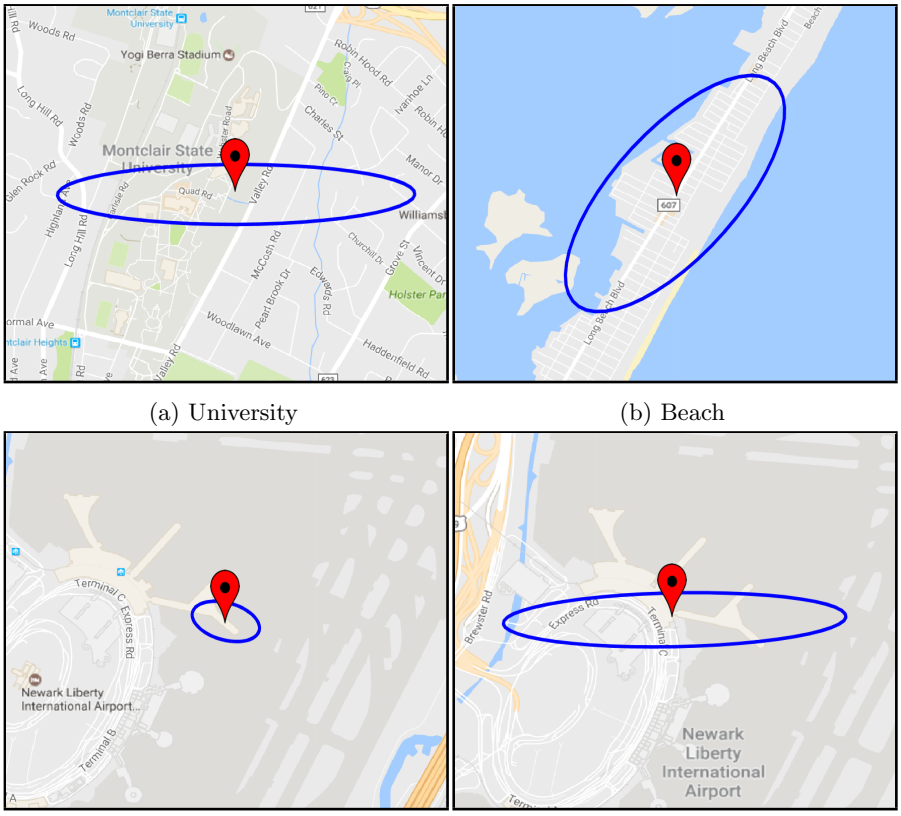
1 Case Study

We show some example latent regions and topics. In particular, we select 100 most informative regions from the 10,000 learned regions ranked by the entropy w.r.t. the topic distribution of each region. Regions with high entropy only mention particular topics rather than covering all topics. We randomly select four regions from the 100 informative regions, and plot them in Figure 1. The top two topics and their representative words (i.e., words with high probability) for each region are listed in Table 1.

Table 1: Topics for each region

Region	Topic	Representative words
University	4879	class, hit, sleep, caring, clothes, easier
	2987	campus, silent, weekend, pic, nights
Beach	612	beach, sweatshirt, remember, eat, hold
	2929	Drinking, drink, #Veterans, cheat, maybe
Airport (lounges)	958	urge, phone, @mangomeagan, Playing, fm
	7955	delayed, #Weather, fan, engaged, destined
Airport (gate)	9321	Uber, ending, #z100MendesMadness, ready, fam
	4428	airport, let, carrier, #cathedral, hashtag

Observe from Figure 1a and Figure 1b, the university area covers topics such as class and campus life, while the topics in beach area focus on entertainment (e.g., wearing, eating, and drinking). In addition, Figure 1c and Figure 1d shows that even in the same airport, the model discovered different topical-regions such as gates and lounges. Topics in lounges are mainly about things to do for time killing and flight delays, while in the gate area, people often talk about arrival/departure and carrier information.



(a) University (b) Beach

(c) Airport - lounges (d) Airport - gate

Figure 1: Example regions learned on the Twitter dataset. The ovals are the contour lines at confidence level $p = 0.9$. The pins point out the centers of the regions.